

Lab 6

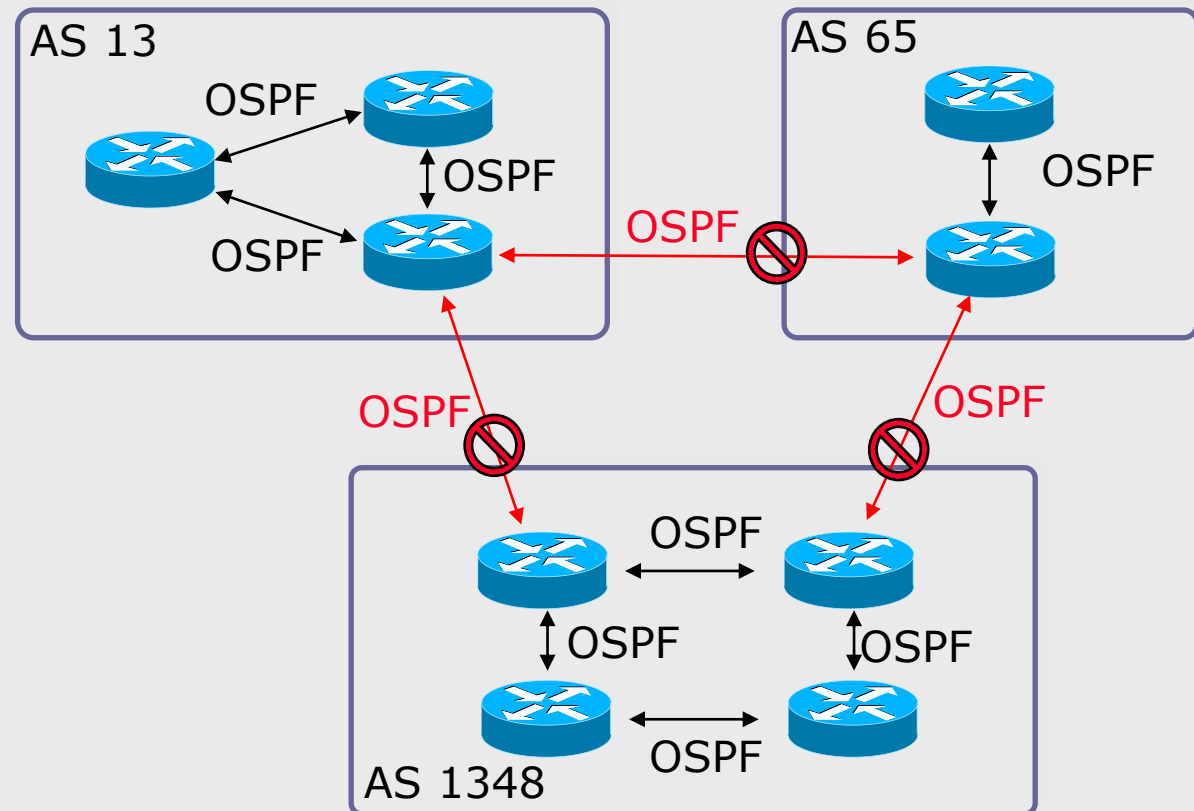
OSPF protocol – single area

OSPF – Open Shortest Path First

- Routing protocol implementing the *link state algorithm*
- Version
 - **OSPF version 2** [RFC 2328]
 - OSPF version 3 [RFC 5340] – for IPv6
- Speed of convergence
- Support for CIDR and VLSM
- Network reachability
 - Virtually no limitations on network size
- Low protocol overhead
- Generic link cost metric
- Careful planning and configuration is needed

OSPF – general principles

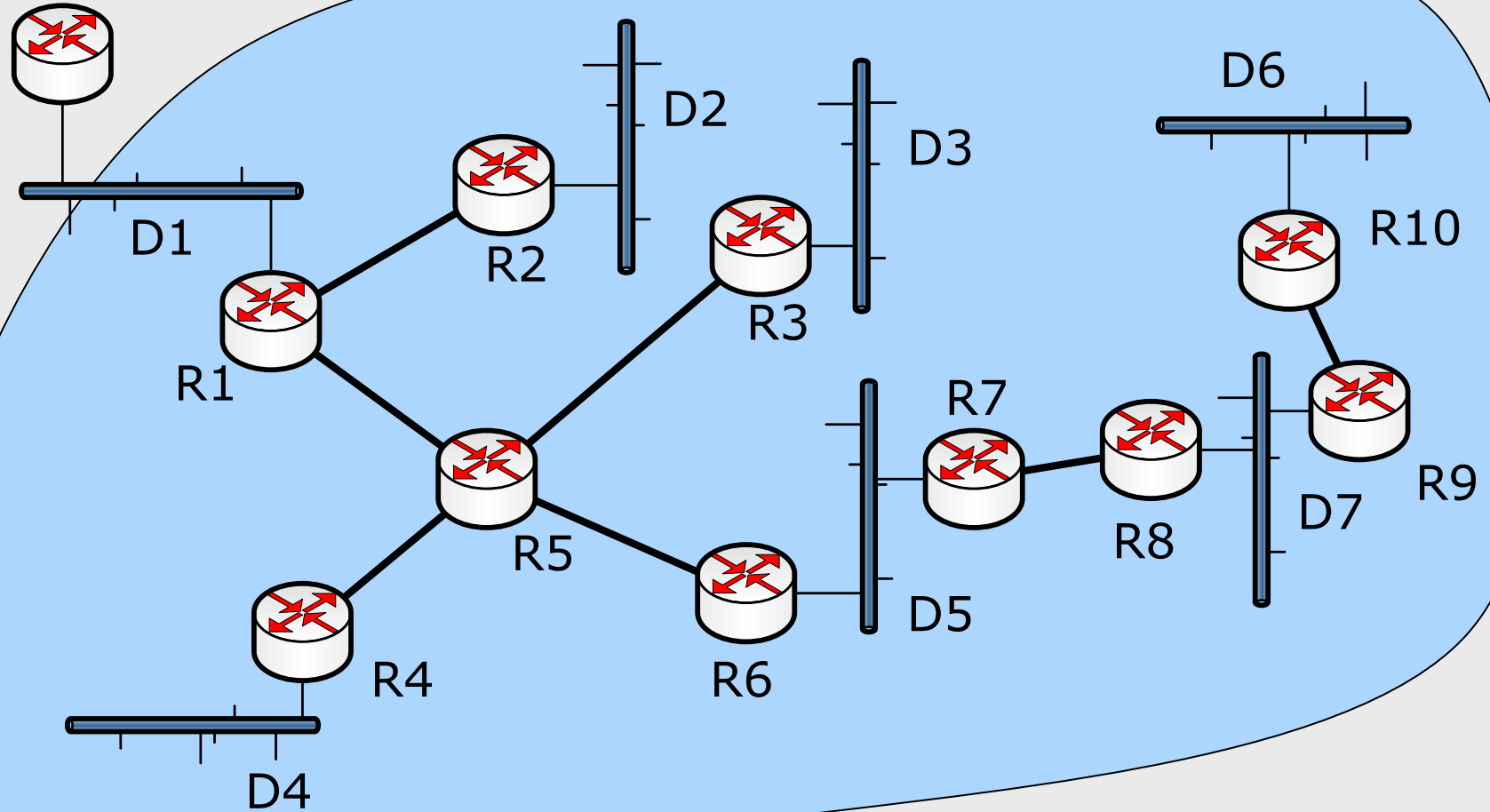
- A single OSPF routing domain is referred to as an *Autonomous System*
 - In the Internet, each AS is uniquely identified by a centrally-assigned number
- Within the AS, each router is uniquely identified by a 32-bit ***router id***



OSPF – general principles

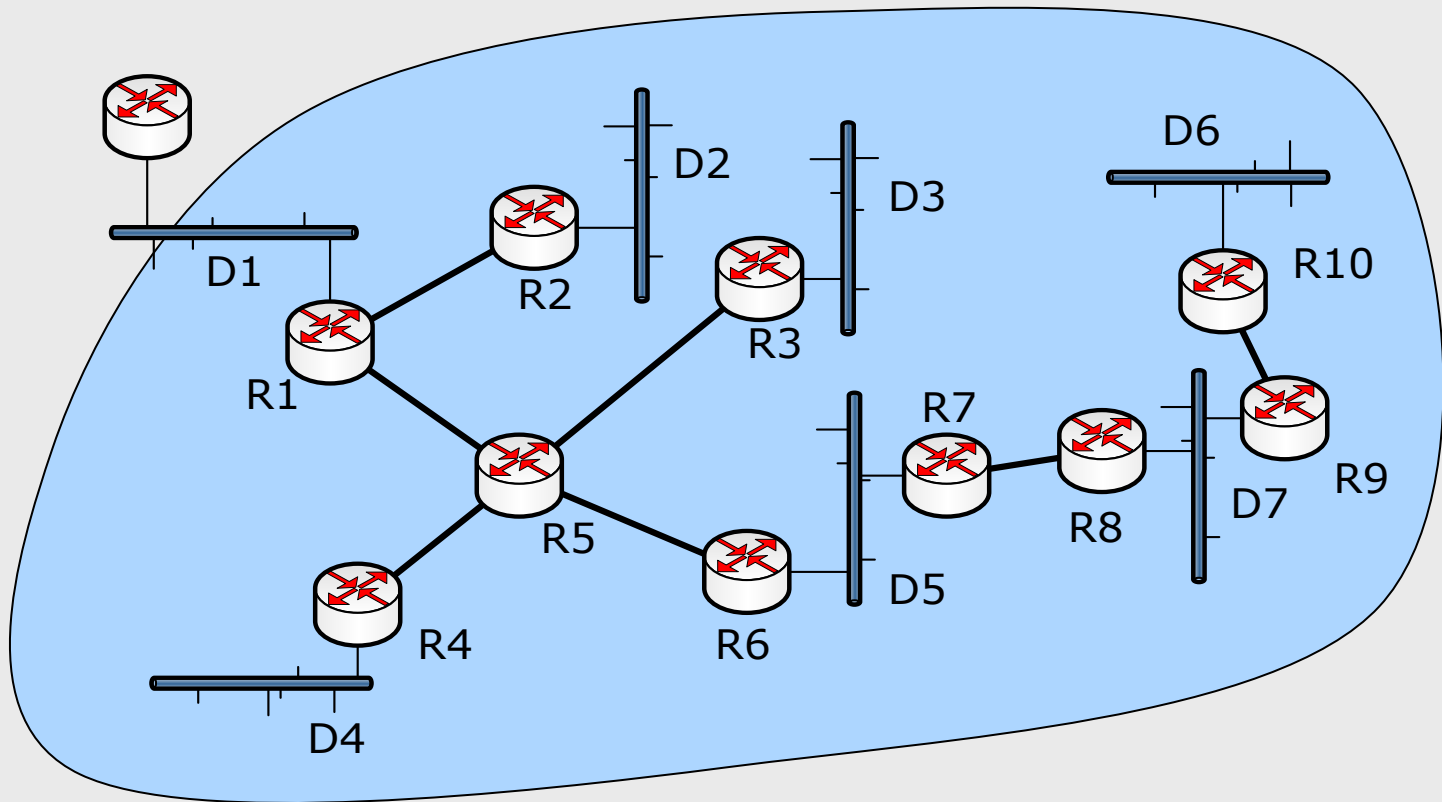
- To support scalability of operation, the AS can be further partitioned into *areas*
 - An area includes a connected subset of links and attached router interfaces
 - A router belongs to an area if it has an interface attached to it
- Each area is identified by a unique 32-bit ***area id***
 - 4-octect dotted notation is used
- OSPF operates in a hierarchical manner
 - Routing is managed **independently** in each area
 - Mechanisms are defined for the exchange of information between areas

OSPF – areas within an AS



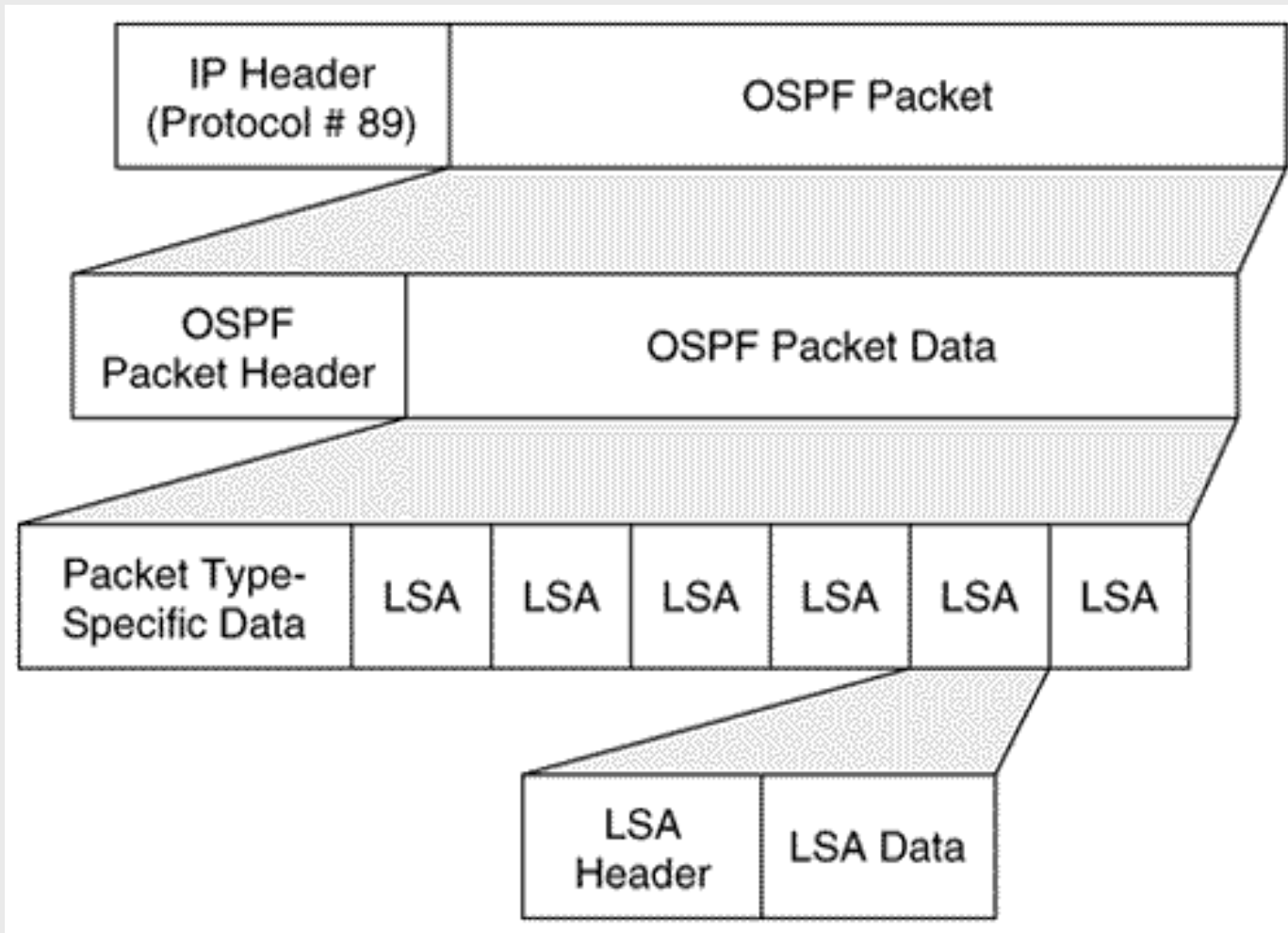
OSPF – router types

- *Internal router* (e.g. R2): all interfaces in the same area
- *AS boundary router* (e.g. R1): at least one interface attached to another AS



OSPF Link State Packets

- Five packet **types** with a common header



OSPF common header

- Version
 - 2
- Router ID
 - ID of the packet's source
- Area ID
 - Area the packet belongs to
- Authentication
- Type
 - 1 – Hello
 - 2 – Database Description
 - 3 – Link State Request
 - 4 – Link State Update
 - 5 – Link State Acknowledgement

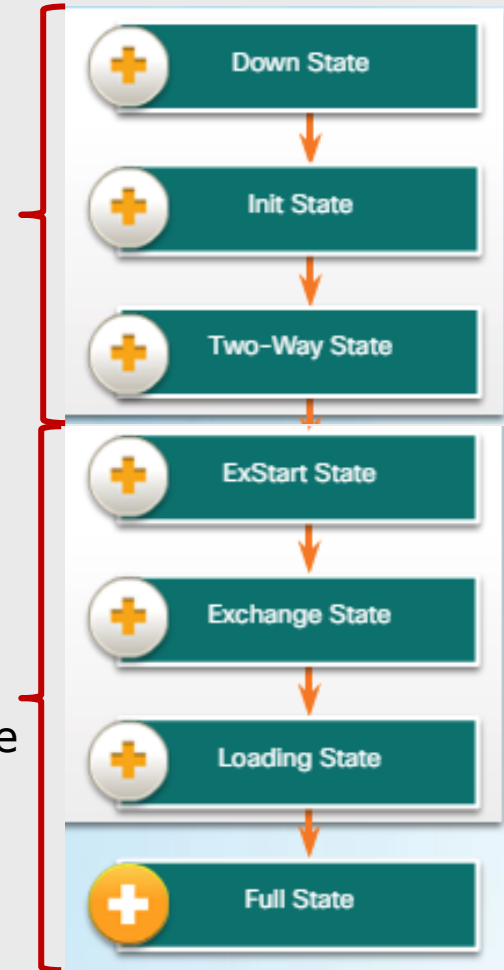
Version	Type	Length
Router ID		
Area ID		
Checksum		AuthType
AuthData		
AuthData		

OSPF Link State Packets

- **Hello** Packet (type 1)
 - provides a means for dynamic neighbor discovery
 - supports the election of Designated and Backup Designated Routers on a LAN segment
- **Database Description** Packet (type 2)
 - All routers within the same area share the same link state database
 - This packet is used to allow quick synchronization between adjacent routers without waiting for LSA flooding
- **Link State Request** Packet (type 3)
 - is sent to ask for a specific set of LSAs to an adjacent router
- **Link State Update** Packet (type 4)
 - is sent either in response to a request or to implement LSA flooding
- **Link State Acknowledgement** Packet (type 5)
 - used to make the flooding of LSAs reliable
 - each LSA received by a router from a neighbor must be explicitly acknowledged

OSPF – protocol operation

- At startup an OSPF router forms **adjacencies** with its neighbor routers over which to pass routing information
- Adjacencies are established in four general phases
 - **Neighbor discovery**
 - Established and maintained through the exchange of *hello packets*
 - *Hellos* are multicast to the *AllSPFRouters* multicast address (224.0.0.5)
 - **Bidirectional communication**
 - This communication is accomplished when two neighbors list each other's Router IDs in their Hello packets
 - **Database synchronization**
 - Packets are exchanged to ensure that both neighbors have identical information in their link-state databases
 - For the purposes of this process, one neighbor will become the master and the other will become the slave
 - **Full adjacency**

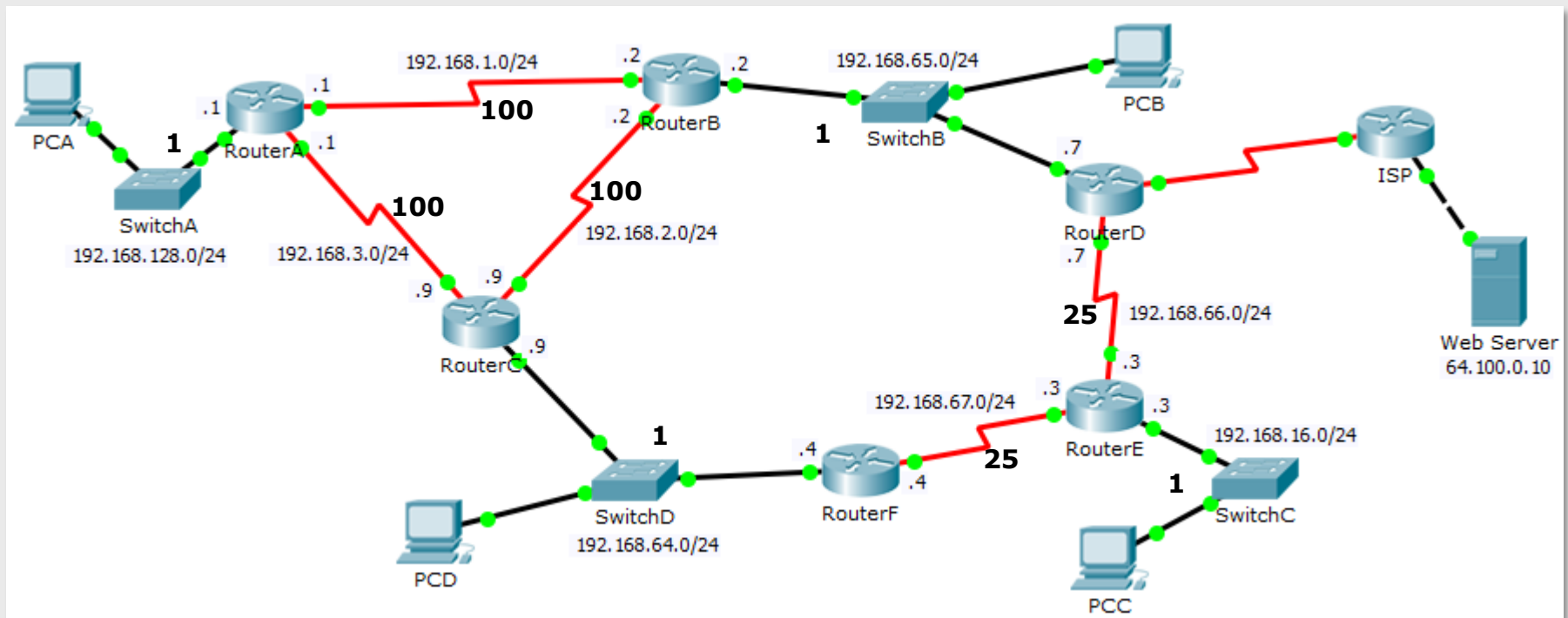


OSPF – protocol operation

- A cost is associated to each interface (referred to as *interface output cost*)
 - Costs are assigned administratively, or
 - Computed by a default (non-standard) rule
- Topological information is exchanged by means of **LSAs** (***Link State Advertisements***)
 - *A router has a separate link-state database for each area it is connected to*

Example network (1)

- Single area – One AS boundary router (RouterD)
- Router IDs: $n.n.n.n$



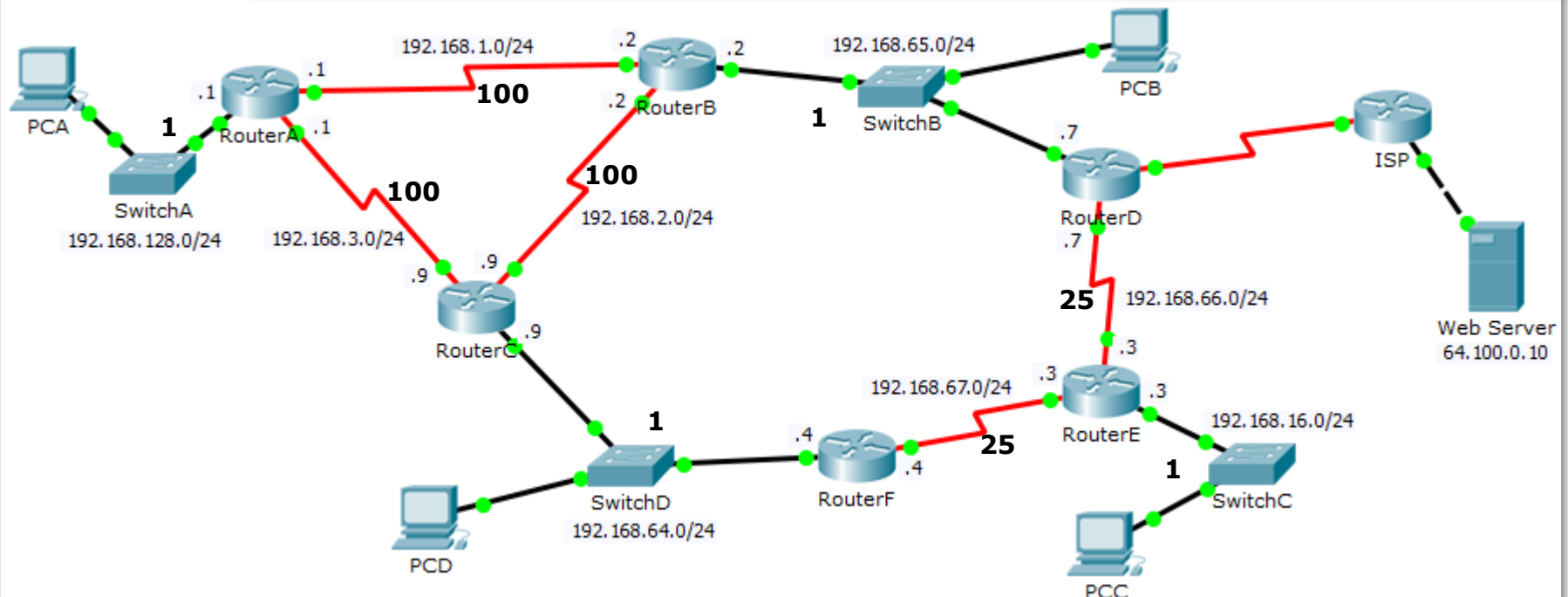
IP rout

```
RouterA#show ip route
```

```
...
```

```
Gateway of last resort is 192.168.1.2 to network 0.0.0.0
```

```
C    192.168.1.0/24 is directly connected, Serial0/0/0
O    192.168.2.0/24 [110/200] via 192.168.1.2, 00:04:53, Serial0/0/0
      [110/200] via 192.168.3.9, 00:04:53, Serial0/0/1
C    192.168.3.0/24 is directly connected, Serial0/0/1
O    192.168.16.0/24 [110/127] via 192.168.1.2, 00:04:18, Serial0/0/0
      [110/127] via 192.168.3.9, 00:04:18, Serial0/0/1
O    192.168.64.0/24 [110/101] via 192.168.3.9, 00:04:18, Serial0/0/1
O    192.168.65.0/24 [110/101] via 192.168.1.2, 00:04:18, Serial0/0/0
O    192.168.66.0/24 [110/126] via 192.168.1.2, 00:04:18, Serial0/0/0
O    192.168.67.0/24 [110/126] via 192.168.3.9, 00:04:18, Serial0/0/1
C    192.168.128.0/24 is directly connected, FastEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 192.168.1.2, 00:04:18, Serial0/0/0
```



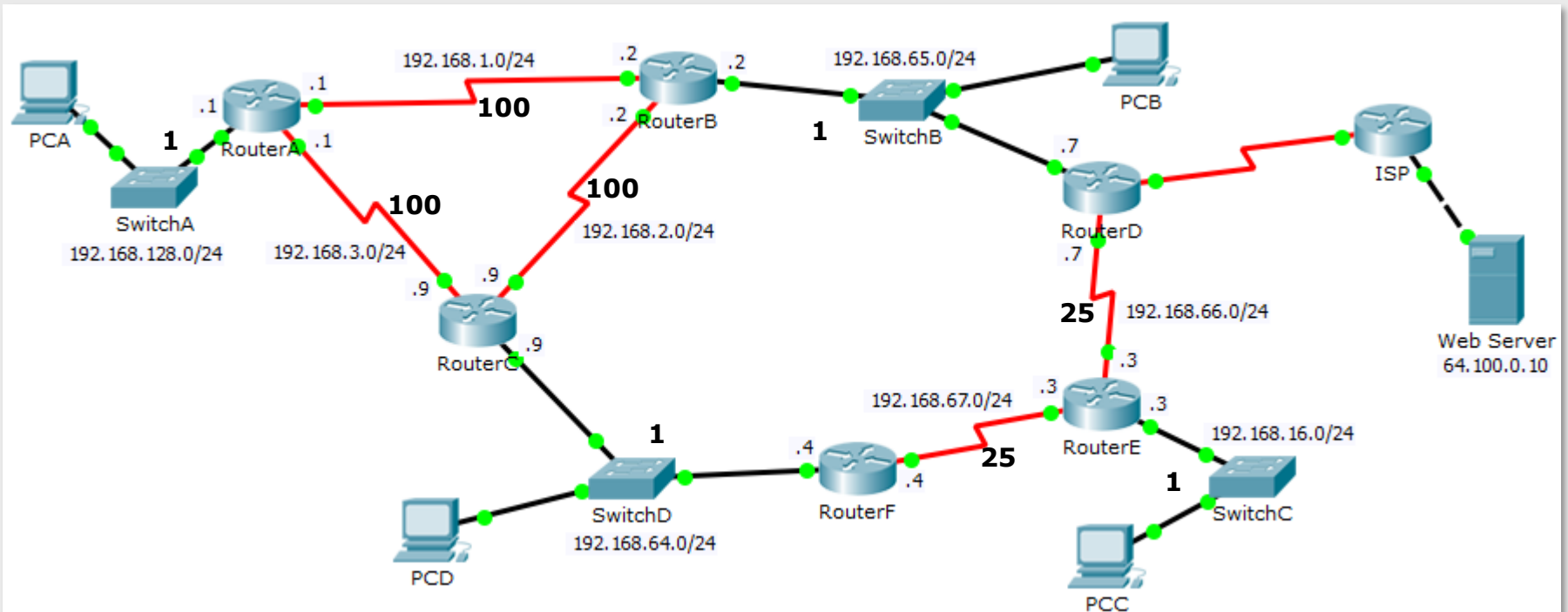
IP rout

```
RouterF#show ip route
```

```
...
```

```
Gateway of last resort is 192.168.67.3 to network 0.0.0.0
```

```
O    192.168.1.0/24 [110/151] via 192.168.67.3, 00:05:52, Serial0/0/0
O    192.168.2.0/24 [110/101] via 192.168.64.9, 00:05:52, FastEthernet0/0
O    192.168.3.0/24 [110/101] via 192.168.64.9, 00:05:52, FastEthernet0/0
O    192.168.16.0/24 [110/26] via 192.168.67.3, 00:06:32, Serial0/0/0
C    192.168.64.0/24 is directly connected, FastEthernet0/0
O    192.168.65.0/24 [110/51] via 192.168.67.3, 00:05:52, Serial0/0/0
O    192.168.66.0/24 [110/50] via 192.168.67.3, 00:06:32, Serial0/0/0
C    192.168.67.0/24 is directly connected, Serial0/0/0
O    192.168.128.0/24 [110/102] via 192.168.64.9, 00:05:52, FastEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 192.168.67.3, 00:06:32, Serial0/0/0
```



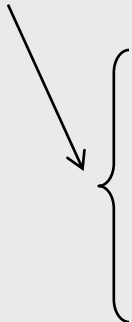
Hello Packet

Version	1	Length	
Router ID			
Area ID			
Checksum		AuthType	
AuthData			
AuthData			
Network Mask			
Hello Interval		Options	Priority
Router Dead Interval			
Designated Router			
Backup Designated Router			
Neighbor			
Neighbor			

**OSPF
Packet
Header**

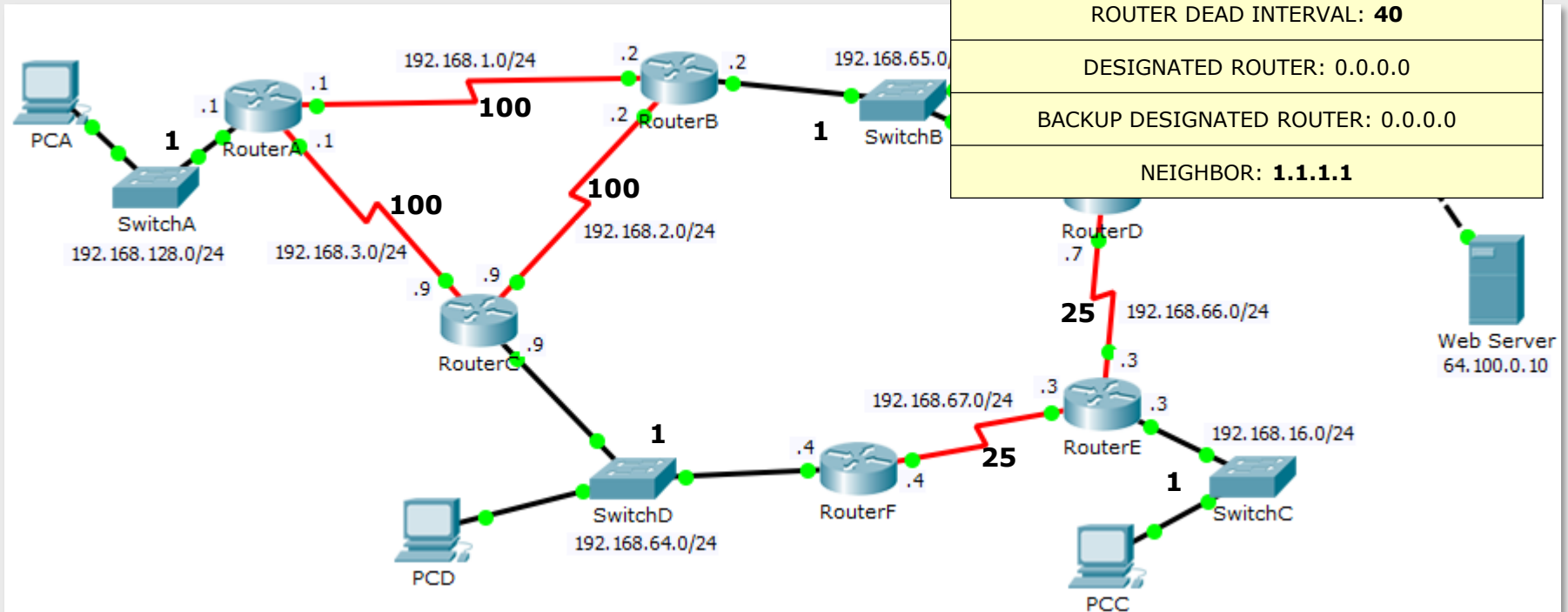
**Hello
Packet
Format**

**List of
Neighbors**



Hello Packet

■ RouterC on Serial Se0/0/0



Link State Update Packet

- It includes one or more **Link State Advertisements**
- The LSA might have been generated by a router other than the one sending the packet (LSA flooding)

Version	4	Length
Router ID		
Area ID		
Checksum	AuthType	
AuthData		
AuthData		
Number of LSAs		
LSA Bodies		

Link State Advertisements

1	Router LSA
2	Network LSA
3	ABR Summary LSA
4	ASBR Summary LSA
5	AS External Route LSA
6	Multicast Group LSA
7	NSSA External LSA
9	Opaque LSA: Link-Local Scope
10	Opaque LSA: Area-Local Scope
11	Opaque LSA: AS Scope

Link State Age	Options	LS Type
Link State Identification		
Advertising Router		
Link State Sequence Number		
Link State Checksum	Length	

1	Hello Packet
2	Database Description Packet
3	Link State Request Packet
4	Link State Update Packet
5	Link State Acknowledgement Packet

Version	Type = 4	Packet Length
Router Identification		
Area Identification		
Checksum	Authentication Type	
Authentication Data		
Authentication Data		
Number of LSAs		
Link State Header		
Link State Data		

OSPF
Packet
Header

LSA
Bodies

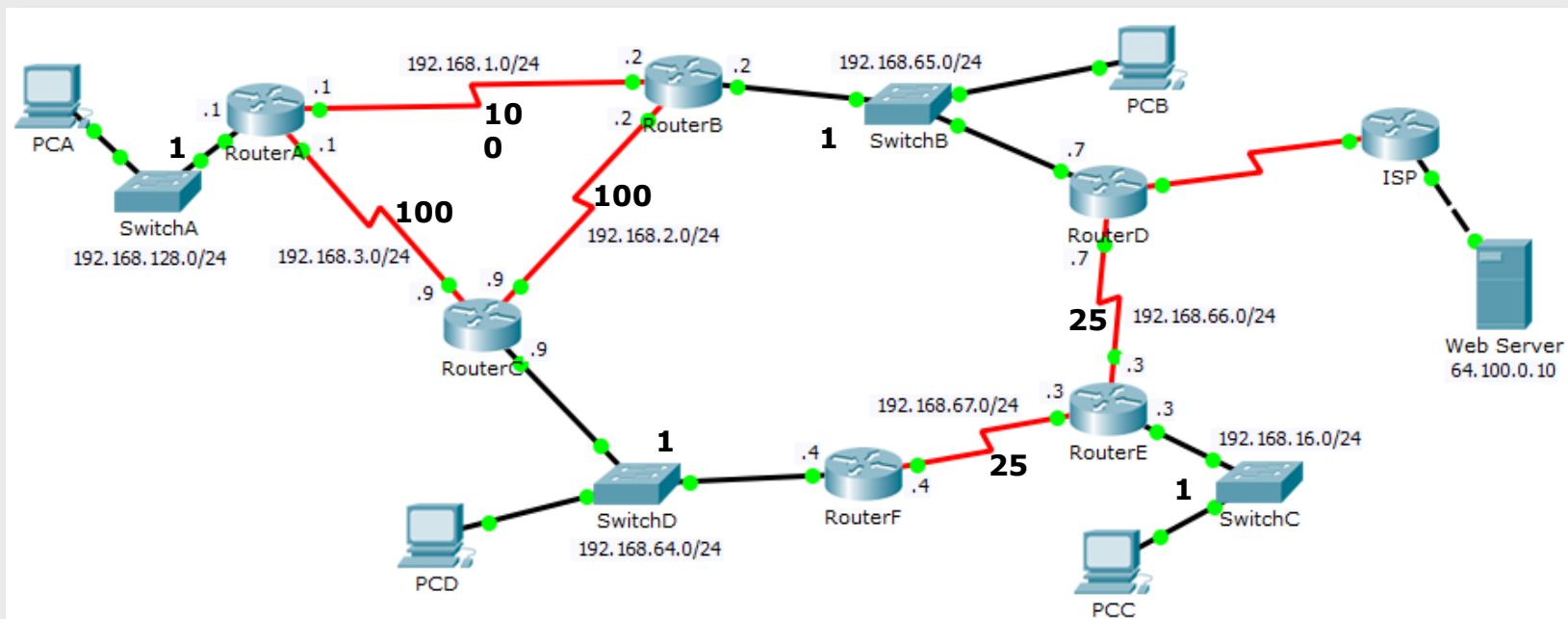
Link State Header

Link State Age	Options	LS Type
Link State Identification		
Advertising Router		
Link State Sequence Number		
Link State Checksum	Length	

- Aging mechanism to enforce quick topology updates
 - LS IDentification interpretation depends on the LS type
 - Advertising Router: the router that generated the LSA
 - LS Sequence Number: to enable selective flooding
-
- An LSA is uniquely identified by its LS Type, LS ID and Advertising Router
 - The most recent instance is determined by examining the LS age and the LS Sequence Number

Router links LSA (LS type 1)

- It includes adjacency information of a router, and related costs
- Is propagated by selective flooding only within the area where it has been generated
 - Internal routers specifies the full adjacency information
 - ABRs specify the subset of adjacencies within the area
 - One Router LSA per area is thus generated
 - ABRs flood a Router LSA only through the interfaces belonging to the same area from which it has been received



Router links LSA (LS type 1)

Link State Age		Options	1
Link State Identification			←
Advertising Router			
Link State Sequence Number			
Link State Checksum		Length	
Rtype	0	Number of Links	
Link ID			
Link Data			
Type	#TOS	TOS 0 Metric	
TOS	0	Metric	
TOS	0	Metric	

is the Router ID of the router that generated the LSA

Version	Type = 4	Packet Length
Router Identification		
Area Identification		
Checksum		Authentication Type
Authentication Data		
Authentication Data		
Number of LSAs		
Link State Header		
Link State Data		

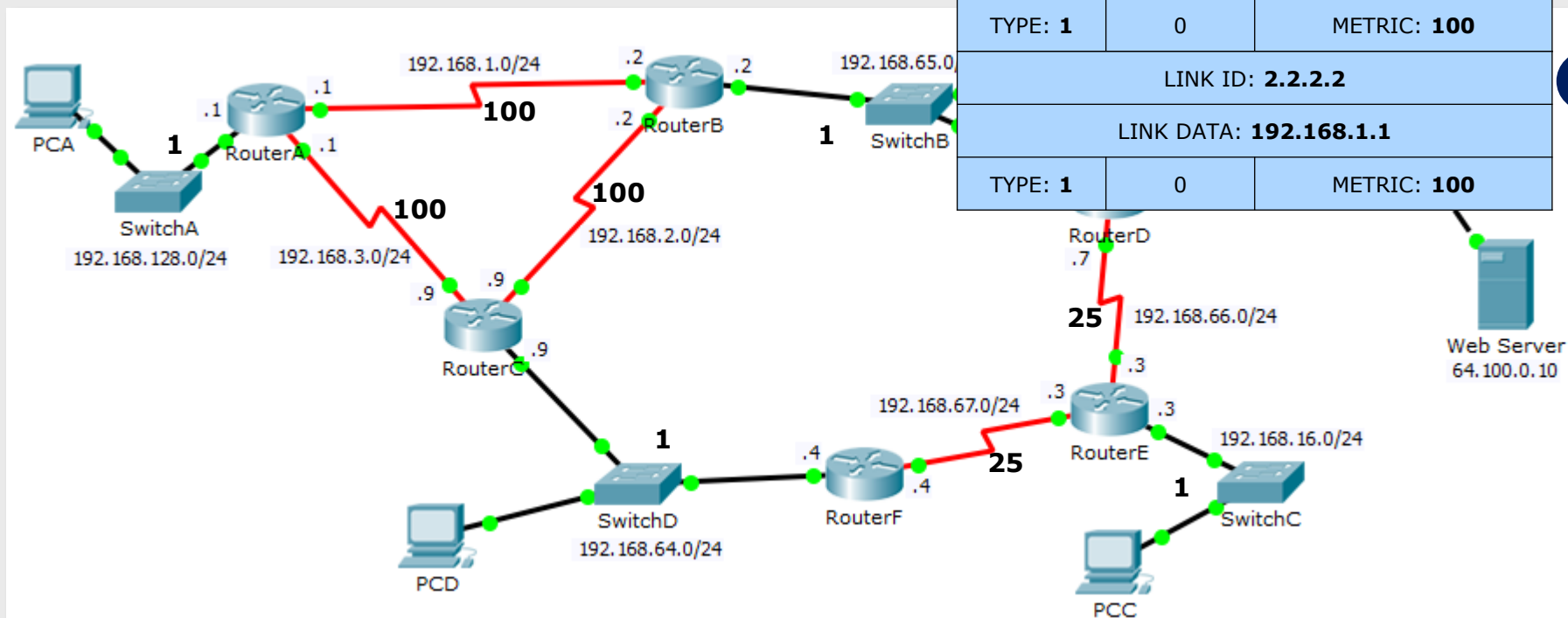
OSPF
Packet
Header

LSA
Bodies

Router links LSA (LS type 1)

■ RouterA

- Link type 1 – point-to-point connection to another router

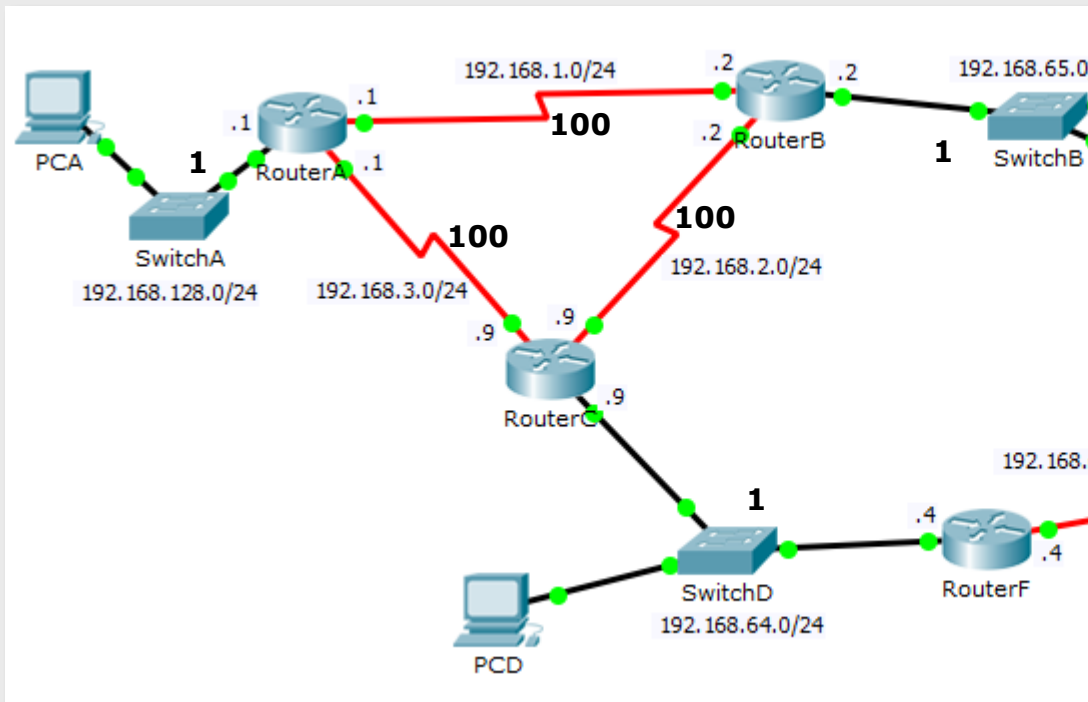


LSA AGE: 0		OPTIONS: 0	LS TYPE: 1
LINK STATE ID: 1.1.1.1			
ADVERTSING ROUTER: 1.1.1.1			
LS SEQUENCE NUM: 0X80000006			
CHECKSUM: 27683		LENGTH: 84	
V+E+B	0	LINK COUNT: 5	
LINK ID: 9.9.9.9			
LINK DATA: 192.168.3.1			
TYPE: 1	0	METRIC: 100	
LINK ID: 2.2.2.2			
LINK DATA: 192.168.1.1			
TYPE: 1	0	METRIC: 100	

Router links LSA (LS type 1)

■ RouterA

- Link type 3 – stub network



LSA AGE: 0		OPTIONS: 0	LS TYPE: 1
LINK STATE ID: 1.1.1.1			
ADVERTSING ROUTER: 1.1.1.1			
LS SEQUENCE NUM: 0X80000006			
CHECKSUM: 27683		LENGTH: 84	
V+E+B	0	LINK COUNT: 5	
LINK ID: 192.168.128.0			
LINK DATA: 255.255.255.0			
TYPE: 3	0	METRIC: 1	
LINK ID: 192.168.1.0			
LINK DATA: 255.255.255.0			
TYPE: 3	0	METRIC: 100	
LINK ID: 192.168.3.0			
LINK DATA: 255.255.255.0			
TYPE: 3	0	METRIC: 100	

OSPF database

OSPF Router with ID (4.4.4.4) (Process ID 1)

Router Link States (Area 0)

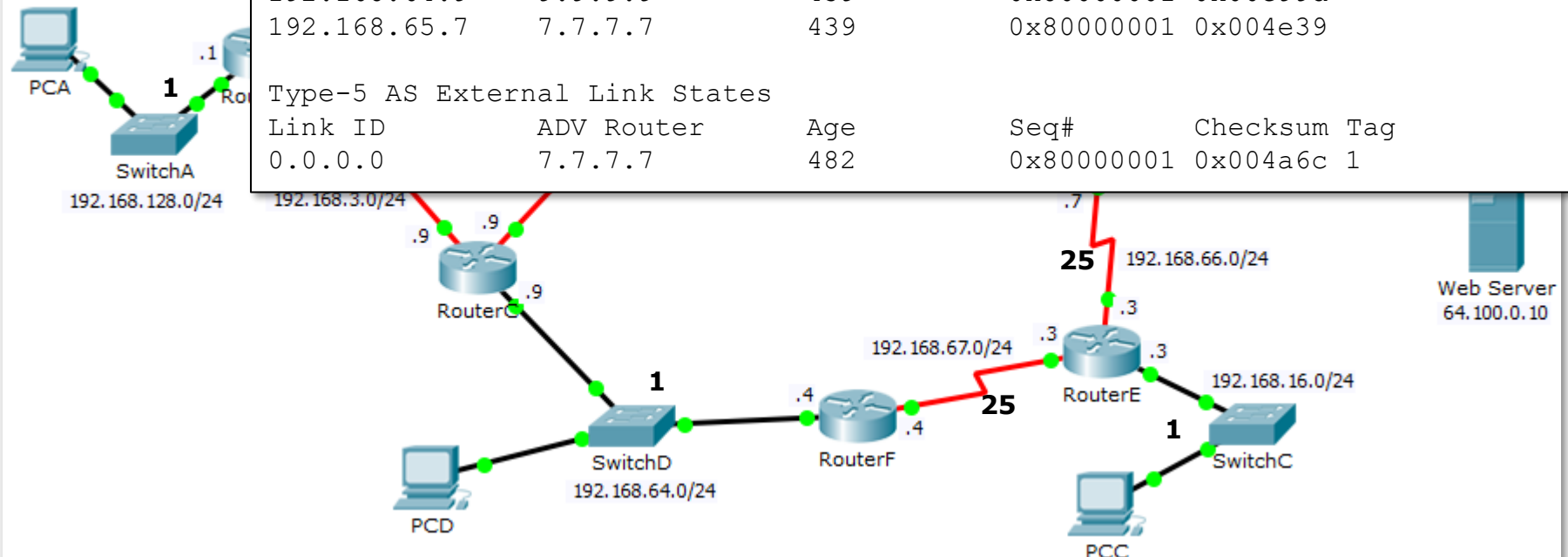
Link ID	ADV Router	Age	Seq#	Checksum	Link count
3.3.3.3	3.3.3.3	474	0x80000006	0x006637	5
1.1.1.1	1.1.1.1	473	0x80000006	0x006fd4	5
4.4.4.4	4.4.4.4	439	0x80000005	0x005c71	3
7.7.7.7	7.7.7.7	439	0x80000006	0x00edbb	3
2.2.2.2	2.2.2.2	438	0x80000007	0x004b80	5
9.9.9.9	9.9.9.9	434	0x80000007	0x000b90	5

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.64.9	9.9.9.9	439	0x80000001	0x00e99d
192.168.65.7	7.7.7.7	439	0x80000001	0x004e39

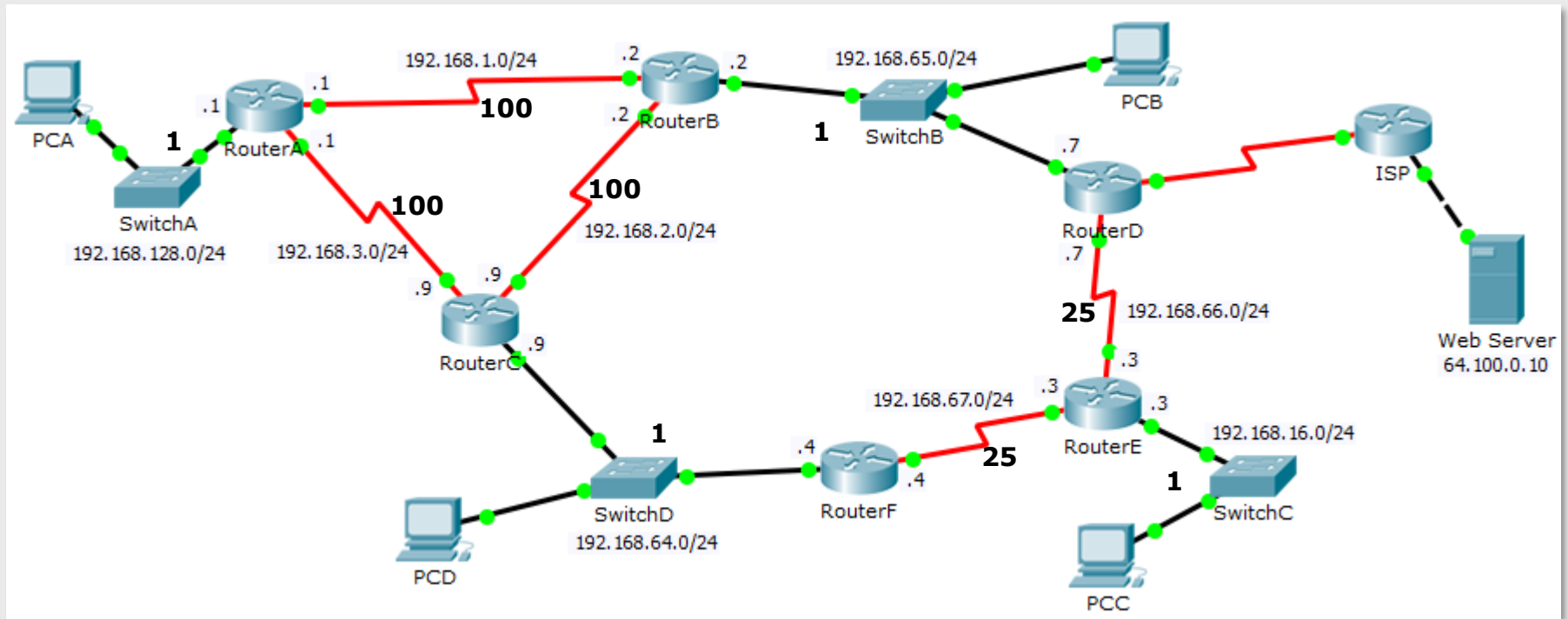
Type-5 AS External Link States

Link ID	ADV Router	Age	Seq#	Checksum	Tag
0.0.0.0	7.7.7.7	482	0x80000001	0x004a6c	1



AS external link LSA (LS type 5)

- Generated by ASBRs to advertise networks outside the AS
- Propagated by selective flooding throughout the AS irrespectively of area boundaries



AS external link LSA (LS type 5)

Link State Age		Options	5	
Link State Identification			←	
Advertising Router				
Link State Sequence Number				
Link State Checksum		Length		
Network Mask				↙
E	0	Metric		
Forwarding Address				
Route Tag				
E	TOS	TOS Metric		
Forwarding Address				
Route Tag				

is the advertised external network address

is the advertised external network mask

Version	Type = 4	Packet Length
Router Identification		
Area Identification		
Checksum		Authentication Type
Authentication Data		
Authentication Data		
Link State Header		
Link State Data		

OSPF
Packet
Header

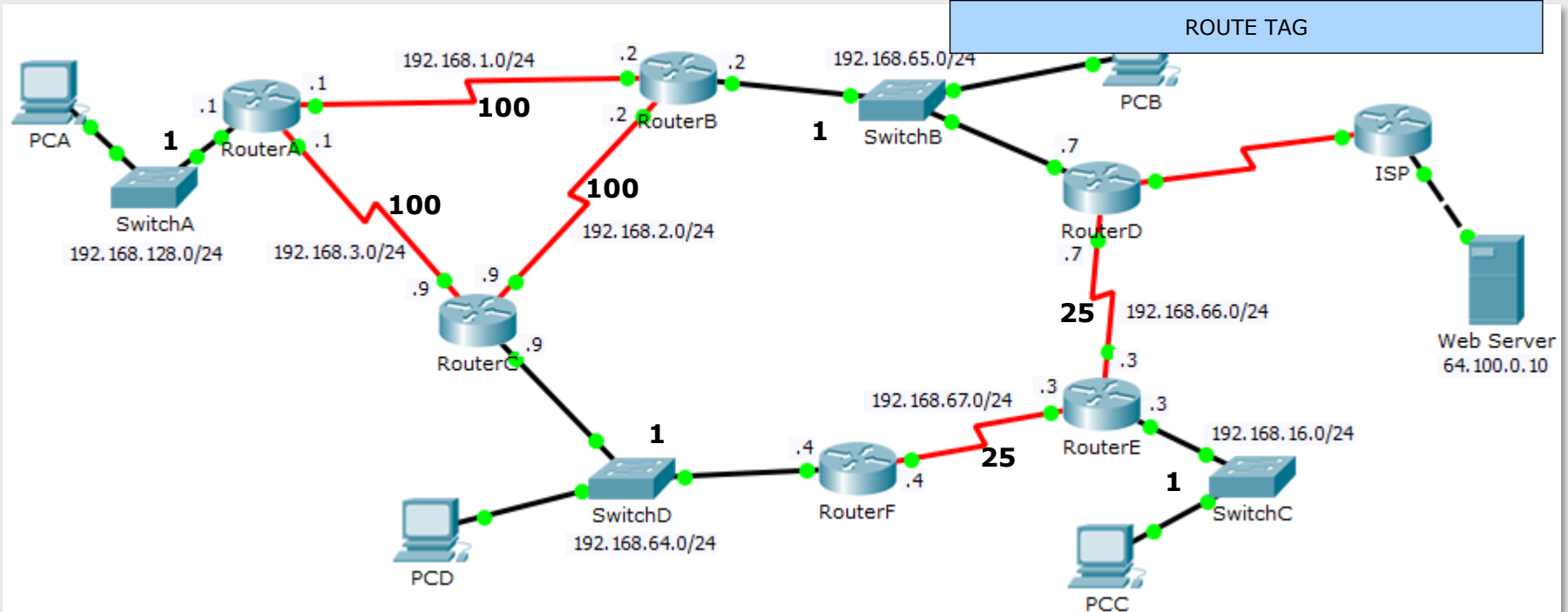
LSA
Bodies

AS external link LSA (LS type 5)

■ RouterD

- Default route for the AS

LSA AGE: 44		OPTIONS: 0	LS TYPE: 5
LINK STATE ID: 0.0.0.0			
ADVERTISING ROUTER: 7.7.7.7			
LS SEQUENCE NUM: 0X80000001			
CHECKSUM: 19052		LENGTH: 36	
NETWORK MASK: 0.0.0.0			
E: 1	0	METRIC: 1	
FORWARDING ADDRESS: 0.0.0.0			
ROUTE TAG			



OSPF database

```
OSPF Router with ID (4.4.4.4) (Process ID 1)
```

Router Link States (Area 0)

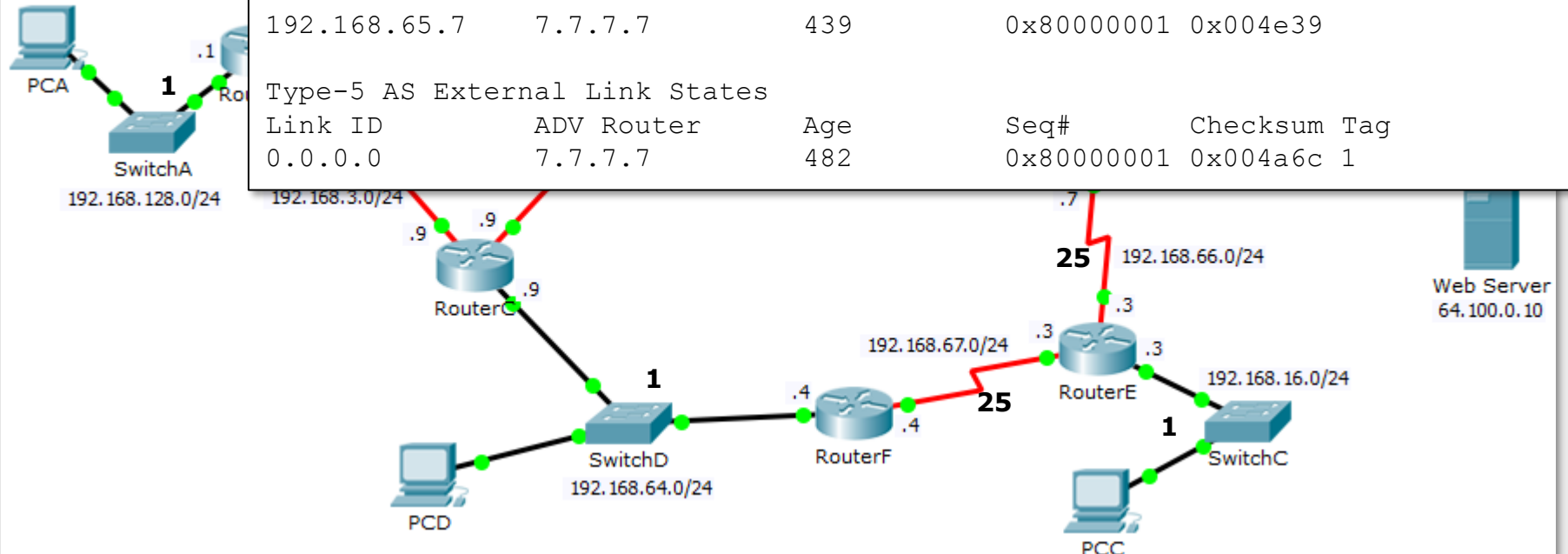
Link ID	ADV Router	Age	Seq#	Checksum	Link count
3.3.3.3	3.3.3.3	474	0x80000006	0x006637	5
1.1.1.1	1.1.1.1	473	0x80000006	0x006fd4	5
4.4.4.4	4.4.4.4	439	0x80000005	0x005c71	3
7.7.7.7	7.7.7.7	439	0x80000006	0x00edbb	3
2.2.2.2	2.2.2.2	438	0x80000007	0x004b80	5
9.9.9.9	9.9.9.9	434	0x80000007	0x000b90	5

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.64.9	9.9.9.9	439	0x80000001	0x00e99d
192.168.65.7	7.7.7.7	439	0x80000001	0x004e39

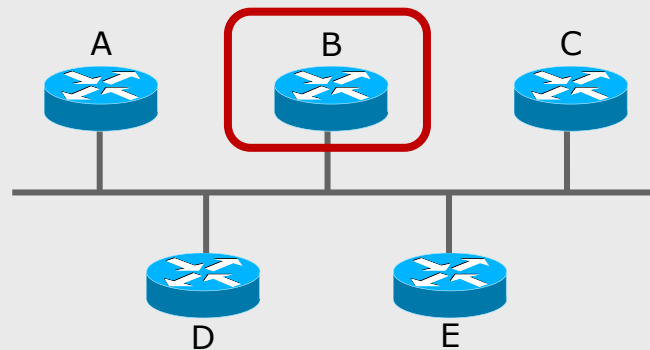
Type-5 AS External Link States

Link ID	ADV Router	Age	Seq#	Checksum	Tag
0.0.0.0	7.7.7.7	482	0x80000001	0x004a6c	1



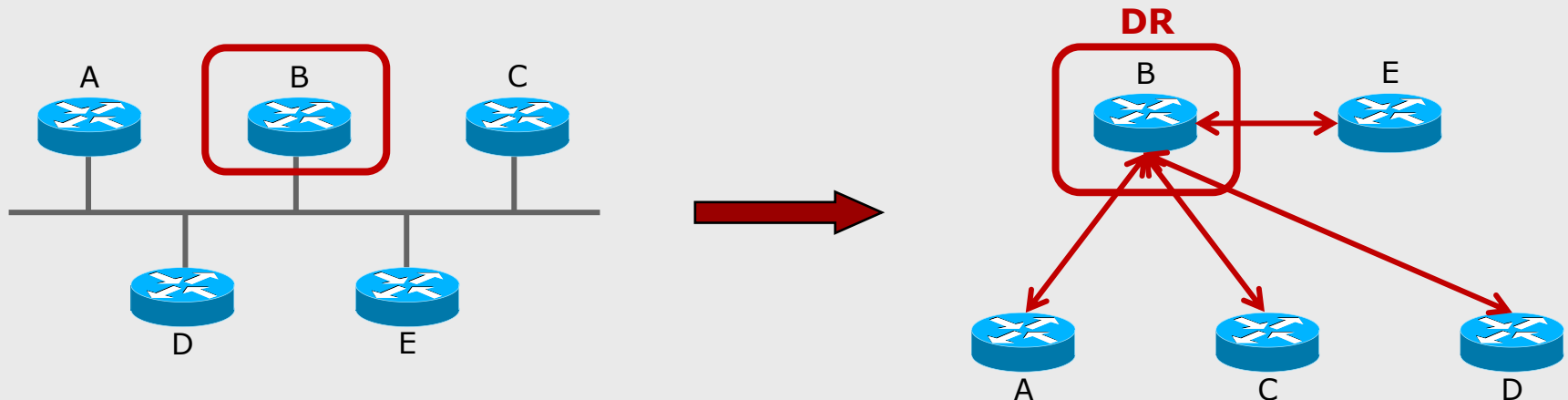
OSPF – broadcast networks

- Broadcast networks (LAN, MAN) deserve special treatment because of their *any-to-any* flexibility
- Multiple routers may be adjacent pairwise on the same link
- Example: five routers on the same LAN
 - Since each router can communicate with the other four, each would consider that it had four neighbors
 - Five routers, each with four neighbors, creates a total of 20 (type-1 link) entries in the link state database
 - Moreover, LSAs are flooded and “reflooded” on the same LAN



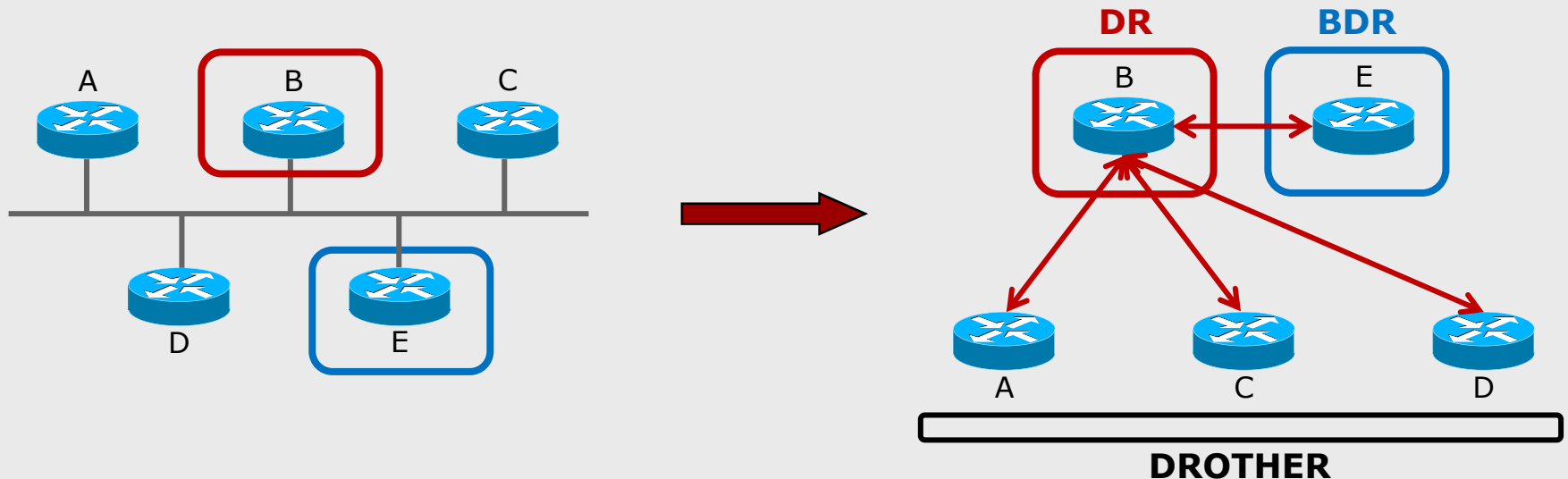
OSPF – broadcast networks

- To help keep the size of link state databases manageable, OSPF treats broadcast networks a special way
- It elects a special router from among those present on the network (based on a configured *router priority*)
- This router, known as the **Designated Router** (DR), treats all routers on the network as adjacent neighbors, whereas the other routers consider the DR as their only(*) adjacent neighbor



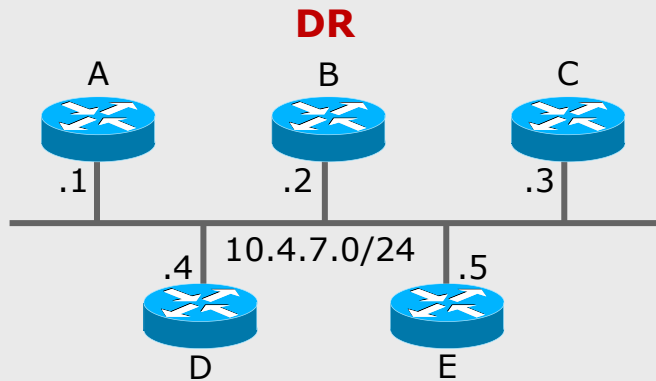
OSPF – broadcast networks

- It elects a special router from among those present on the network (based on a configured *router priority*)
- This router, known as the **Designated Router** (DR), treats all routers on the network as adjacent neighbors, whereas the other routers consider the DR as their only(*) adjacent neighbor
- (*)
 - A **Backup DR** (BDR) is also elected
 - Dedicated *AllDRouters* multicast address **224.0.0.6**



OSPF – broadcast networks

- How is the broadcast network described (in the link state DB)?
- All routers: add a **transit link** (type 2) to their own Router LSA, pointing to the DR, with corresponding cost
- DR: send a **Network LSA** (type 2), listing all the routers (IDs) attached to the broadcast network
- #entries in the link state DB now increase *linearly* with the number of routers!



A
Router LSA
10.4.7.2 1

B
Router LSA
10.4.7.2 1

C
Router LSA
10.4.7.2 1

D
Router LSA
10.4.7.2 1

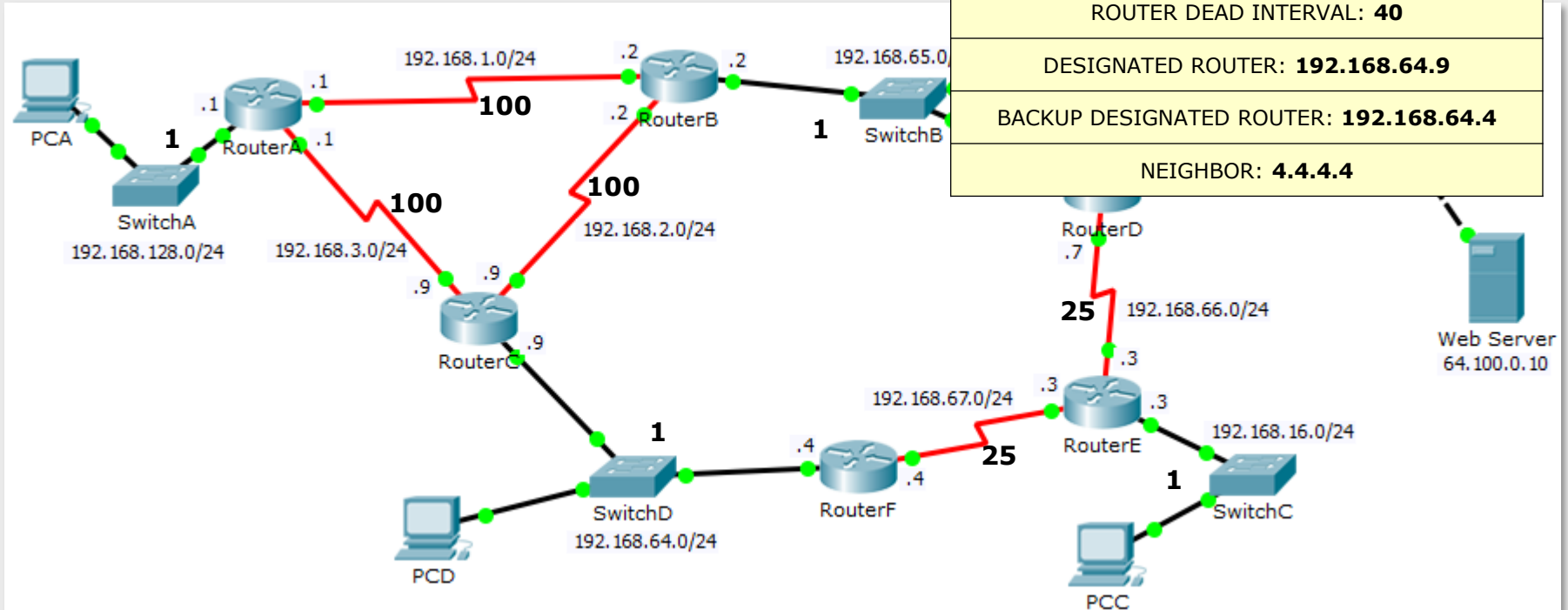
E
Router LSA
10.4.7.2 1



B
Network LSA
10.4.7.0/24
A
B
C
D
E

Hello Packet

■ RouterC on Fa0/0

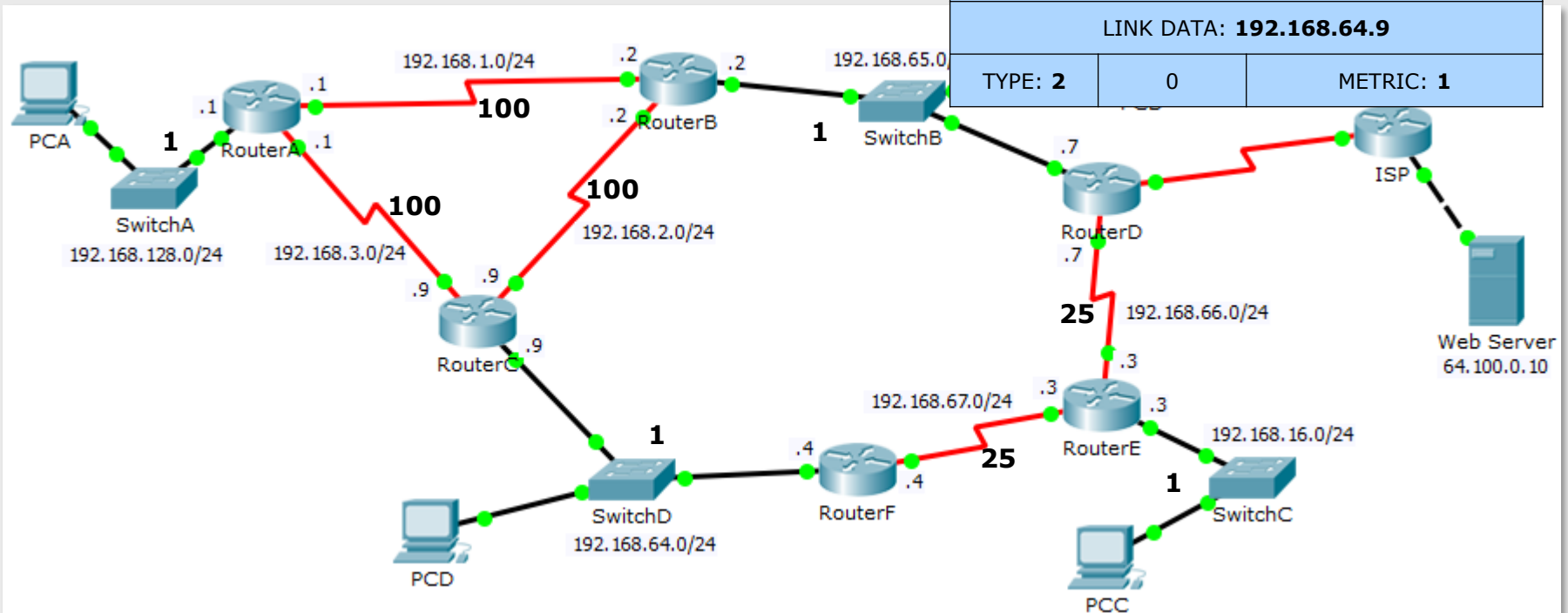


Router links LSA (LS type 1)

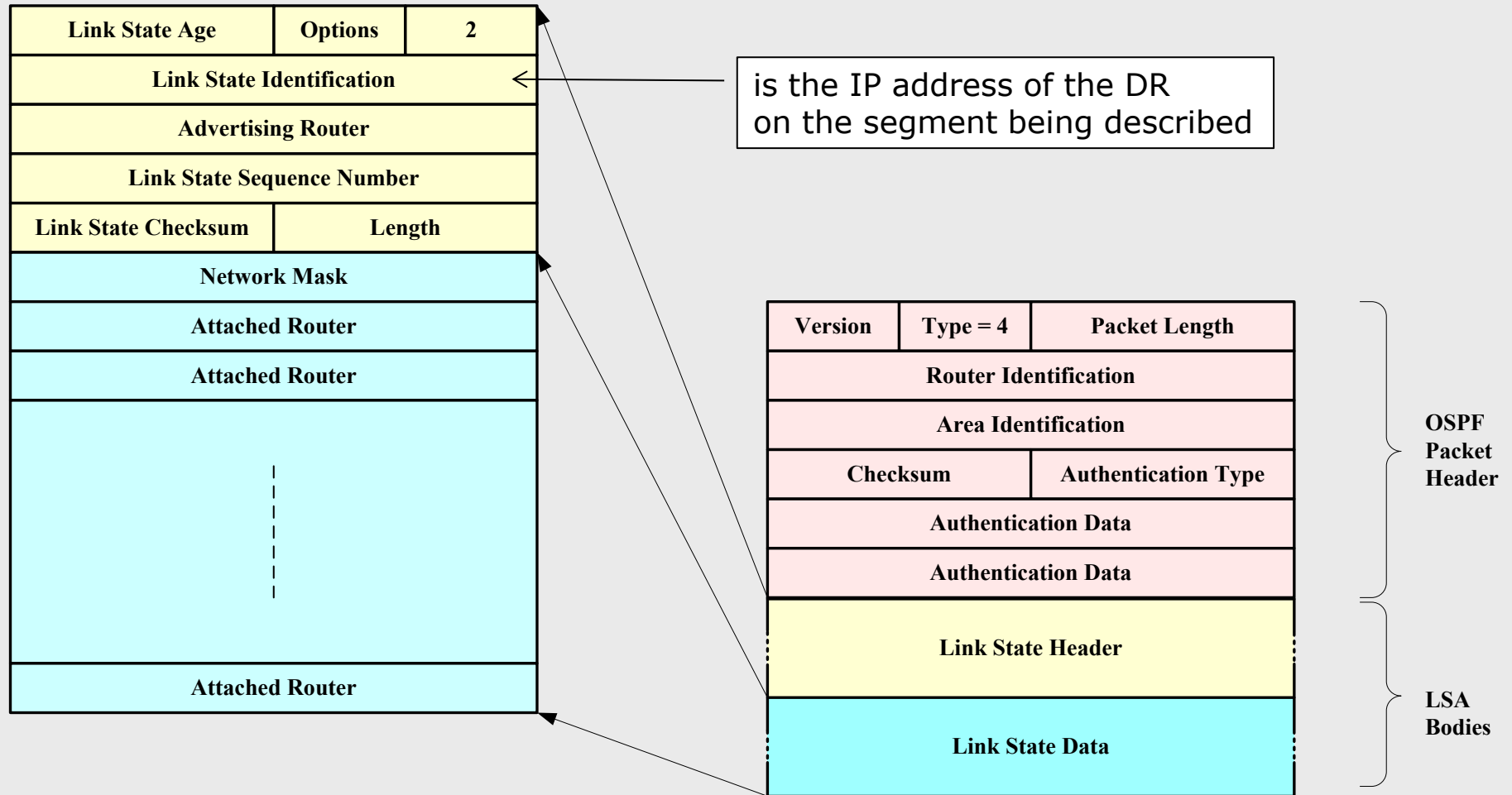
■ RouterC

- Link type 2 – transit network

LSA AGE: 0		OPTIONS: 0	LS TYPE: 1
LINK STATE ID: 9.9.9.9			
ADVERTSING ROUTER: 9.9.9.9			
LS SEQUENCE NUM: 0X80000007			
CHECKSUM: 15705		LENGTH: 84	
V+E+B	0	LINK COUNT: 5	
LINK ID: 192.168.64.9			
LINK DATA: 192.168.64.9			
TYPE: 2	0	METRIC: 1	



Network links LSA (type 2)

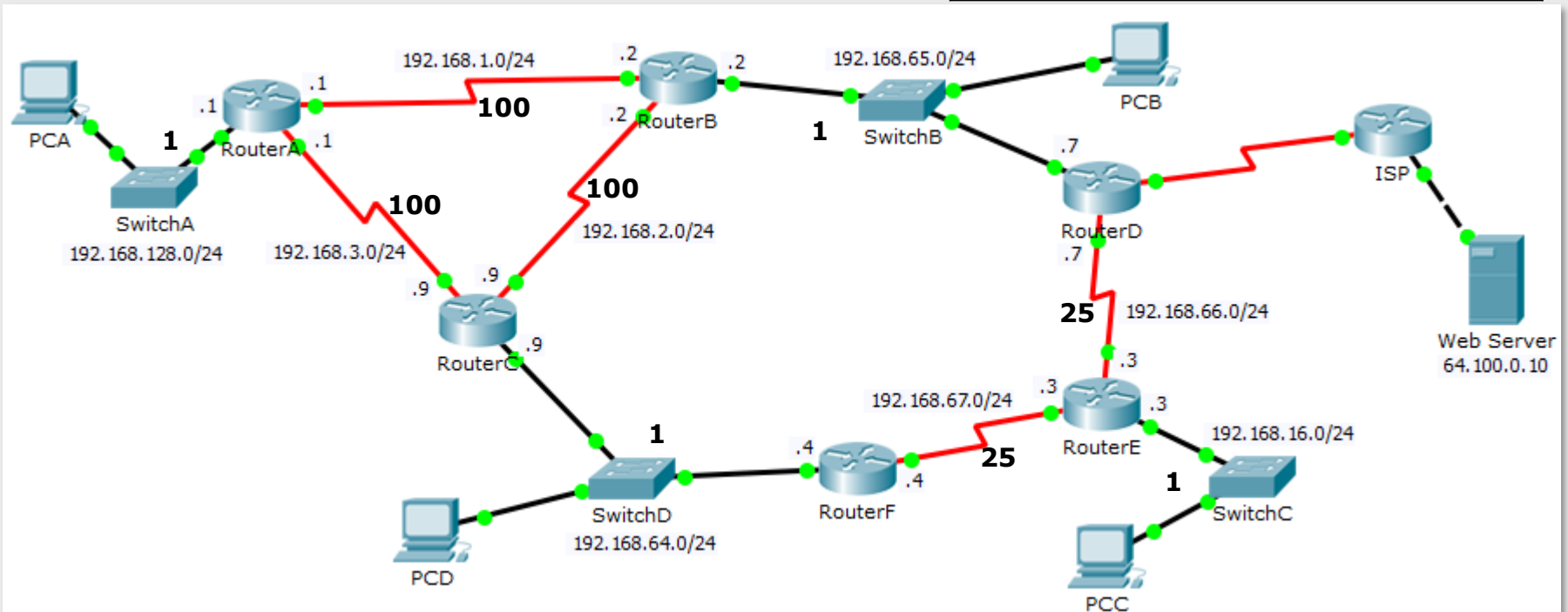


Network links LSA (LS type 2)

■ RouterC

- Designated router on LAN D

LSA AGE: 0	OPTIONS: 0	LS TYPE: 2
LINK STATE ID: 192.168.64.9		
ADVERTSING ROUTER: 9.9.9.9		
LS SEQUENCE NUM: 0X80000001		
CHECKSUM: 45610	LENGTH: 32	
NETWORK MASK: 255.255.255.0		
ATTACHED ROUTER: 4.4.4.4		
ATTACHED ROUTER: 9.9.9.9		



OSPF database

OSPF Router with ID (4.4.4.4) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
3.3.3.3	3.3.3.3	474	0x80000006	0x006637	5
1.1.1.1	1.1.1.1	473	0x80000006	0x006fd4	5
4.4.4.4	4.4.4.4	439	0x80000005	0x005c71	3
7.7.7.7	7.7.7.7	439	0x80000006	0x00edbb	3
2.2.2.2	2.2.2.2	438	0x80000007	0x004b80	5
9.9.9.9	9.9.9.9	434	0x80000007	0x000b90	5

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.64.9	9.9.9.9	439	0x80000001	0x00e99d
192.168.65.7	7.7.7.7	439	0x80000001	0x004e39

Type-5 AS External Link States

Link ID	ADV Router	Age	Seq#	Checksum	Tag
0.0.0.0	7.7.7.7	482	0x80000001	0x004a6c	1

